1 Background

Reducing the volume of inductors and transformers is an important route to increasing power density in power converters [1]. Their soft magnetic cores guide magnetic flux and influence electromagnetic performance [2]. The magnetic flux distribution within a core depends on the geometry, material properties and operating conditions. Nonuniform flux density can lead to uneven core losses and localised heating [3]. In some inductor designs, regions near the conductor approach magnetic saturation while material farther away remains underutilised [4]. Near saturation, further increases in magnetising field produce relatively little additional flux density as the material's incremental permeability decreases. This can reduce the effective inductance as the current increases, limiting the utilisation of the core [2, 5]. These limitations motivate designs that distribute magnetic flux more effectively.

One approach is to optimise the core geometry and winding arrangement. For example, Cui et al. designed an inductor with an improved flux distribution by shaping the core and distributing the winding turns [6]. A complementary approach is to vary permeability spatially within the core to tailor the flux distribution [4, 7]. Wang et al. experimentally demonstrated that a three-permeability inductor achieved higher inductance than a single-permeability counterpart and improved the efficiency of the tested high-frequency DC/DC converter [4]. Liu et al. reported that a three-permeability toroidal core achieved the same inductance at a smaller volume than a single-permeability core [7]. These results demonstrate the potential benefits of spatial permeability variation for specific designs, although the benefits depend on the core geometry and operating conditions. Geometry and permeability are not mutually exclusive design variables, and optimising both may offer further benefits [8].

Several manufacturing approaches have been investigated to realise spatially varying permeability. Lu et al. produced discrete radial permeability regions by compacting individual rings at different pressures [9]. Continuous radial permeability variation has also been demonstrated by imposing a temperature gradient during sintering [10]. These approaches demonstrate the feasibility of creating multipermeability cores, but the reported implementations rely on specific core geometries and manufacturing arrangements.

Additive manufacturing offers an alternative for producing customised geometries and positioning different material formulations within a component. Several AM processes have been investigated for soft-magnetic materials, for example powder-bed fusion, binder jetting and material extrusion [11, 12]. The suitability of each process depends on the feedstock, required magnetic properties and post-processing conditions. Material extrusion is particularly relevant for thermoplastic polymers and composites because it offers relatively low initial investment, simple operation and the capacity for simultaneous use of multiple materials [13].

Liu et al. demonstrated that material extrusion can be used to manufacture multipermeability cores. They employed a multi-extruder paste-based 3D printer to fabricate a toroidal core with three distinct permeability regions [7]. Their ferrite paste required sintering above 900 °C, whereas their powdered-iron paste could be cured below 250 °C. This demonstrates the feasibility of additive manufacturing for multipermeability cores, while illustrating the importance of feedstock formulation and post-processing requirements.

Within material extrusion, the choice of feedstock involves a trade-off between magnetic performance and manufacturability. Producing filament-based composites with high magnetic-particle loading while retaining suitable processability is challenging [14]. Cañada et al. report that filament-based soft-magnetic cores in the studies they reviewed generally exhibited relative permeabilities below 10. Paste-based approaches can achieve much higher permeabilities, but their rheology can complicate dimensional accuracy, and some formulations require high-temperature sintering. In contrast, pellet-based feedstock can provide moderate permeability

without high-temperature post-processing. Cañada et al. demonstrated nylon-12/FeSiAl composite cores with a relative permeability of approximately 42 using pellet-based material extrusion [15]. Since these results come from different materials and studies, they do not constitute a controlled comparison of the three feedstock types. Nevertheless, pellet extrusion provides a promising starting point for this work because it has already demonstrated useful soft-magnetic properties without high-temperature post-processing.

The composition of a polymer-based magnetic composite influences its magnetic properties. Hu et al. experimentally investigated the permeability of polymer composites with different magnetic inclusion contents, although their matrix differed from the nylon-12 composite considered here [16]. Huber et al. demonstrated variable-composition printing by blending NdFeB-filled PA12 filament with unfilled PA12 filament. Their approach controlled the remanent magnetisation distribution of hard-magnetic permanent magnets to produce a predefined stray field, rather than the effective permeability of a soft-magnetic core [17]. This demonstrates that magnetic composition can be varied during material extrusion, but does not establish how pellet blend ratio affects the permeability of a soft-magnetic composite.

Composition-controlled pellet extrusion has also been investigated in non-magnetic material systems. Brackett et al. studied in-situ blending of neat and carbon-fibre-filled ABS pellets and showed that transitions between compositions occupy a finite distance and depend on the direction of the material change [18]. This indicates that changes in the input blend ratio may not translate immediately into corresponding changes in the extruded material. However, the material system and large-format extrusion process they investigated differ from the present application. Elsersawy et al. found that dedicated mixing elements improved the uniformity of blended and coloured PLA compared with a standard Mahor V4 pellet extruder without a mixing element [19]. Although this demonstrates the importance of mixing configuration, colour uniformity does not establish uniform magnetic filler distribution or reproducible permeability. It therefore remains unclear whether a standard pellet extruder provides sufficient mixing to produce predictable and repeatable magnetic properties when blending magnetic-filled and unfilled polymer pellets.

This work will investigate the relationship between pellet blend ratio and the effective permeability of the printed material. It will also assess whether a standard pellet extruder produces sufficiently homogeneous and repeatable material to establish a reliable relationship between nominal blend ratio and measured permeability. If the results indicate that mixing is inadequate, the need for a dedicated mixing element will be investigated. If the relationship between blend ratio and permeability is sufficiently controllable, subsequent work can investigate varying the feed ratio during printing to produce distinct permeability regions within a core.

2 Problem Statement

Previous studies have demonstrated that the magnetic properties of additively manufactured materials can be tailored through composition and that pellet-based material extrusion can produce soft-magnetic polymer composites. However, it remains unclear whether blending magnetic-filled and unfilled polymer pellets provides a reliable means of controlling the effective permeability of the resulting material. The nominal blend ratio may not translate directly into a uniform composition after extrusion, and variations in mixing may affect the consistency of the measured permeability. Establishing a predictable and repeatable relationship between blend ratio and permeability is therefore necessary to determine whether this approach is suitable for controlling magnetic properties during printing. If sufficient control can be achieved, varying the feed composition during printing could provide a route to manufacturing cores with designed permeability distributions.

2.1 Research Question

To address the discovered problems, the work will investigate the following research questions:

Can varying the blend ratio of magnetic-filled and unfilled polymer pellets produce reproducible changes in the permeability of a soft-magnetic composite manufactured using a commercial pellet extruder?

Does a standard pellet extruder produce sufficiently homogeneous and repeatable material to establish a reliable relationship between nominal blend ratio and measured permeability?

Can discrete permeability regions be fabricated by sequentially printing pre-blended pellet batches, using controlled purging to achieve distinct and reproducible material compositions?

2.2 Requirements

To address the research question and capture the overarching scientific plan, the following requirements have been defined:

- Assess whether mixing-related material variations at the five selected nominal blend ratios are sufficiently small to permit meaningful and repeatable permeability characterisation.
- Quantify the effective permeability and between-specimen variability of the printed material at five magnetic-filled-to-unfilled pellet mass ratios, using three independently printed specimens per ratio.
- Establish a repeatable pause–purge–resume procedure for switching between pre-blended pellet batches while limiting residual material from the previous batch.
- Evaluate the feasibility of fabricating a core with three distinct permeability regions by sequentially printing pre-blended pellet batches.

2.3 Delimitations

The project shall use the available commercial pellet extruder and the selected magnetic-filled and unfilled polymer feedstocks as the baseline material-processing system. The development of an automated, continuously adjustable multi-material feeding system is outside the initial scope.

3 Methodology

3.1 Requirements

To address this research question, the following requirements have to be met:

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